

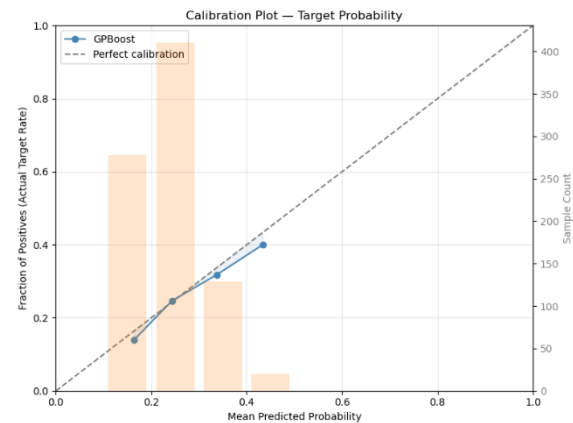
Baltimore Ravens Data Scientist Application Project

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The Problem: The ability of a Cornerback to blanket a Wide Receiver in Man-to-Man coverage and prevent targets entirely is an incredibly useful skill. However, evaluating this skill is difficult, because traditional metrics do not account for the quality of receiver faced, or contextual information such as the route run. This project aims to separate and evaluate **the ability of the cornerback to prevent targets in man-to-man coverage**, accounting for influences outside their control.

The Solution: I developed a model to evaluate how often a WR was targeted on a route when matched up in Man-to-Man coverage against a particular cornerback. The model accounts for receiver quality, route difficulty, and play-scheme context to isolate the corner's individual contribution.

Accuracy: The final model is well calibrated on unseen data. Even on brand new unseen plays, when the model predicts a 25% target rate, roughly 25% of these plays actually result in a target. Additionally, many top corners in the model also appear atop end-of-year PFF player rankings, indicating realistic scores.



Findings: CB skill percentiles for top players and Raven's players are reported below

Top CBs Leaguewide	
Marlon Humphrey	100
D.J. Reed	99.4
Jonathan Jones	98.9

Ravens CBs	
Marlon Humphrey	100
Marcus Peters	89.8
Damarion Williams	27.1

Improvements: This model improves upon traditional CB coverage metrics like raw targets per route run by accounting for the context of each play. Corners that cover more difficult receivers or routes are not penalized; instead, their abilities are adjusted according to the difficulty of their assignments.

Limitations: The largest limitation is the small sample size of man-coverage routes available for each corner. Due to the prevalence of zone coverages around the league, as well as the 9-game sample, very few cornerbacks in the dataset have sufficient data to draw robust conclusions. Supplementing our data with additional games, seasons, or practice data could open the door to potentially stronger evaluations.

Recommended Actions: Scheme Marlon Humphrey to play more man-to-man coverage snaps against high target-earning WRs. The data indicates he may be able to effectively disrupt the offense and reduce target volume towards these players.